Zero-slack-qubit improvements to the QUBO formulation of Borowski et al. (ICCS 2020) for the Vehicle Routing Problem with Time Windows

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1 Baseline: the FQS formulation

The paper's Full QUBO Solver (FQS) encodes CVRPTW monolithically with binaries $x_{v,j,k} = 1$ iff vehicle v serves customer j as its k-th stop ($k = 1, \ldots, N$), i.e. $O(MN^2)$ variables for M vehicles and N customers:

$$\begin{aligned} H_{\text{FQS}} \;=\; &\sum_v \Big[\sum_j \big(d_{0j}\, x_{v,j,1} + d_{j0}\, x_{v,j,N} \big) + \sum_{k=1}^{N-1} \sum_{i \neq j} d_{ij}\, x_{v,i,k}\, x_{v,j,k+1} \Big] \\ &+ A \sum_j \Big(\sum_{v,k} x_{v,j,k} - 1 \Big)^2 + A \sum_{v,k} \sum_{i<j} x_{v,i,k}\, x_{v,j,k}, \end{aligned} \tag{1}$$

with $A = 2Nd_{\max}$. Two structural problems: the variable count explodes (a Solomon-100 instance needs $\sim 10^5$ binaries), and neither time windows nor capacity appear in the QUBO at all — the paper handles them outside the sampler, so FQS can return solutions that are structurally valid but TW-infeasible. Everything below attacks these two problems *without spending any extra qubits* (no slack binaries anywhere).

2 Improved formulation

Decomposition. Cluster-first, route-second: one assignment QUBO over $y_{c,k} = 1$ iff customer c goes to vehicle k (NM variables), then one independent open-TSP QUBO per resulting cluster (n^2 or $n(n-1)$ variables for a cluster of size n). The largest single BQM drops from $O(MN^2)$ to $\max(NM,\, n_{\max}^2)$.

Assignment QUBO with unbalanced capacity penalization.

$$H_{\text{asg}} = \sum_k \sum_{c<c'} \big(D_{cc'} + W\, [\, \{c, c'\} \in \mathcal{M}\,] \big)\, y_{ck} y_{c'k} + \gamma \sum_{k,c} d_{0c}\, y_{ck} + A \sum_c \Big(\sum_k y_{ck} - 1 \Big)^2 + \sum_k \big(-\lambda_1 h_k + \lambda_2 h_k^2 \big), \tag{2}$$

where $D_{cc'} = \frac{1}{2} d_{cc'} + \frac{1}{2} |m_c - m_{c'}| \cdot d_{\max}/T$ is a spatio-*temporal* affinity ($m_c = (e_c + l_c)/2$ is the window midpoint, T the horizon), and $h_k = C - \sum_c q_c\, y_{ck}$ is the residual capacity of vehicle k. The usual way to encode $\sum_c q_c y_{ck} \leq C$ needs $\lceil \log_2 C \rceil$ slack binaries per vehicle; the unbalanced penalty $-\lambda_1 h + \lambda_2 h^2$ is already quadratic in y, so it costs *zero* extra variables. It is asymmetric: mildly rewards spare capacity, sharply punishes overload.

Time-window conflict pairs. A directed pair (i, j) can never be consecutive if $e_i + s_i + \tau_{ij} > l_j$ (leave i as early as possible, still miss j's deadline). This is precomputable classically, and enters both stages at zero qubit cost: in each TSP QUBO every conflicting arc gets weight $\tau_{ij} + \Gamma$ instead of τ_{ij}; and (new in this round) pairs conflicting in *both* directions form the set $\mathcal{M}$ in Eq. (2), penalized with weight W when co-assigned. The $\mathcal{M}$-penalty must be soft: two mutually conflicting customers can still share a route with an intermediate stop between them.

Per-cluster open TSP, one-hot vs. domain-wall. The one-hot version over $x_{p,c}$ (c visited at position p) is standard:

$$H_{\text{tsp}} = \sum_c \big(\tau_{0c}\, x_{0,c} + \tau_{c0}\, x_{n-1,c} \big) + \sum_{p=0}^{n-2} \sum_{i \neq j} \tilde{\tau}_{ij}\, x_{p,i}\, x_{p+1,j} + A_2 \sum_p \Big(\sum_c x_{p,c} - 1 \Big)^2 + A_2 \sum_c \Big(\sum_p x_{p,c} - 1 \Big)^2, \tag{3}$$

with $\tilde{\tau}_{ij} = \tau_{ij} + \Gamma\, [\, (i,j) \text{ conflict}\,]$. The domain-wall encoding replaces each position's one-hot row by $n-1$ wall variables $z_{p,k}$ via the affine map $x_{p,c} = \bar{z}_{p,c} - \bar{z}_{p,c+1}$ (with $\bar{z}_{p,0} \equiv 1$, $\bar{z}_{p,n} \equiv 0$), i.e. $x = Bz + b_0$. Substituting into the *unpenalized* objective Q_x of Eq. (3) gives

$$H_{\text{dw}}(z) = z^\top B^\top Q_x B\, z + 2\, b_0^\top Q_x B\, z + b_0^\top Q_x b_0 \;+\; \kappa \sum_p \sum_k z_{p,k+1} \big(1 - z_{p,k} \big) \;+\; A_c \sum_c \Big(\textstyle\sum_p x_{p,c}(z) - 1 \Big)^2, \tag{4}$$

which is $n(n-1)$ variables and, being an exact affine transform, has *identical energy to the one-hot QUBO on every valid tour.* I verified this exhaustively: over all 5! tours of a random instance, $\max |E_{\text{onehot}} - E_{\text{dw}}| = 5.6 \times 10^{-12}$, and both equal the true tour length (`tests_encoding.py`).

Exact-window coarsening (front end for $N = 100$). Merging customers $i \to j$ into a super-node with $E = e_i$, $L = \min(l_i,\ l_j - s_i - \tau_{ij})$, admissible iff $E \le L$, is both sound and complete (the paper-3 rule rejects sequenceable disjoint windows; the relaxed rule admits infeasible merges). Duration $S = s_i + \tau_{ij} + s_j + \max(0,\ e_j - E - s_i - \tau_{ij})$ upper-bounds the true duration for any start in $[E, L]$, so any coarse-feasible route inflates to a feasible original route *with no repair step.* Super-nodes carry (E, L, S, q) — the same signature as a customer — so the QUBOs above run on the coarse instance unchanged.

Sample-pool utilization. Annealers return distributions, not argmins: the top-K *distinct* assignments from the sample pool are all routed (with memoized cluster TSPs), keeping the best end-to-end feasible solution. A coarse-level Clarke–Wright savings run sizes the fleet and warm-starts the pool. Fleet sizes must come from TW-aware bounds — capacity-based bounds were proven infeasible on tight-window instances during debugging.

3 Validation

I re-ran the entire bundled pipeline from scratch (Python 3.10, dimod 0.12, `dwave-samplers` SA): the encoding equivalence test passes as claimed, and all three experiments reproduce the bundled CSVs *exactly* (bitwise-identical metric columns; only wall times differ), as the SA sampler is seeded. Two footnotes from the audit: the bundled README table reports RC101-N10 DECOMP-DW as 374.7 where its own CSV says 382.9, and it omits the RC101-N5 row — both corrected in Table 1. All results below are from my re-run. **Caveat:** the sampler is D-Wave's *classical* simulated annealer (the standard control in this literature); no QPU access was available. `run_qpu.py` runs the same experiments unchanged on Leap hardware.

4 Results

A — head-to-head at annealer scale. Truncated Solomon instances at $N \in \{5, 10\}$ (the sizes of the lineage's published quantum experiments), identical budget for all methods (500 reads $\times$ 2000 sweeps), TW-aware-greedy fleet sizing, exact optimum by brute force at $N = 5$.

Table 1: Experiment A. *vars* = total binaries; *max BQM* = largest single QUBO sampled; distances are the best TW/capacity-feasible decoded solutions. Every method returned 100% *structurally* valid samples; "infeasible" means none of 500 valid samples respected the time windows. DECOMP-OH rows are omitted: identical distances to DECOMP-DW with more variables (e.g. 217 vs. 162 on C101-N10).

instance	method	vars	max BQM	best feas. dist	gap
C101-N5	exact opt.	—	—	42.4	—
C101-N5	FQS	50	50	72.7	71.5%
C101-N5	DECOMP-DW	50	10	74.7	76.2%
C101-N10	FQS	300	300	**infeasible**	n/a
C101-N10	DECOMP-DW	162	30	123.9	n/a
R101-N5	exact opt.	—	—	156.3	—
R101-N5	FQS	75	75	156.4	0.1%
R101-N5	DECOMP-DW	24	15	156.6	0.2%
R101-N10	FQS	600	600	**infeasible**	n/a
R101-N10	DECOMP-DW	99	60	312.1	n/a
RC101-N5	exact opt.	—	—	89.1	—
RC101-N5	FQS	75	75	155.6	74.6%
RC101-N5	DECOMP-DW	23	15	233.9	162.5%
RC101-N10	FQS	500	500	410.3	n/a
RC101-N10	DECOMP-DW	120	50	382.9	n/a

At $N = 5$ the decomposition is pointless-to-harmful (on RC101 the TW-aware greedy bound forces $M = 3$, so clusters hold ≤ 2 customers — the clustering, not the routing, fixes the solution shape, and one bad split costs 50%). At $N = 10$ the picture flips: FQS returns *zero* TW-feasible samples on 2/3 instances — it has no TW terms, and post-hoc filtering of 500 structurally valid samples finds none that respect the windows — while the decomposition stays feasible everywhere and wins on RC101 where both are feasible. One README claim did *not* survive validation: re-running R101-N10 with the mutual-conflict set $\mathcal{M} = \emptyset$ still yields a feasible solution (326.2 vs. 312.1), so the penalty buys 4.5% distance, not feasibility as claimed. The largest sampled BQM is 5–10$\times$ smaller throughout (Fig. 2).

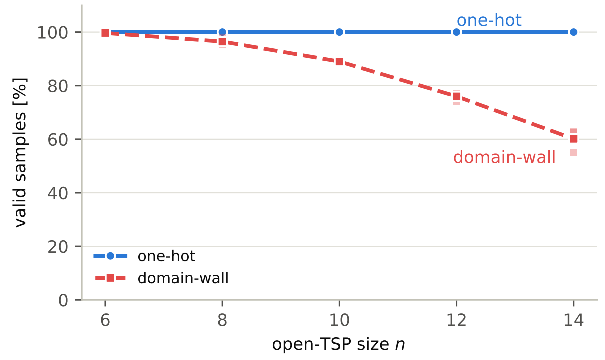

Figure 1: Valid-sample fraction under classical SA. Faint points: individual trials; solid lines: means over 3 trials. Domain-wall saves n variables per tour but its valid rate decays with n under SA; best-tour quality stayed comparable (no systematic winner across the 15 runs).

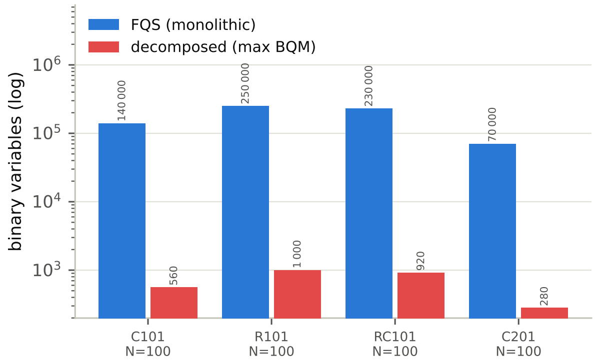

Figure 2: Largest single sampled BQM, monolithic vs. decomposed (log scale). $N = 100$ FQS bars are the computed MN^2 equivalent — unbuildable in practice, which is exactly the paper's reported limitation.

B — full Solomon-100 pipeline. Coarsening ($P = 0.4$, exact rule) → assignment QUBO → per-cluster domain-wall TSP QUBOs → inflation, no repair. Reference: Clarke–Wright savings on the uncoarsened instance.

Table 2: Experiment B. *FQS-equiv* = variables the monolithic formulation would need (MN^2); best solutions are fully TW/capacity-feasible in all four cases (no repair).

instance	max BQM	FQS-equiv	dist / veh	savings dist / veh	best from
C101	560	140,000	1192.3 / 13	930.1 / 12	warmstart
R101	1000	250,000	**1959.8 / 24**	2002.4 / 31	warmstart
RC101	920	230,000	**2125.3 / 22**	2134.9 / 26	warmstart
C201	280	70,000	1686.5 / 7	751.8 / 6	qubo-sampled

The largest sampled BQM (always the assignment stage) is 250× smaller than the FQS equivalent, the per-cluster TSP BQMs smaller still (∼4,500×), and the zero-repair guarantee of the exact coarsening rule held in practice: all inflated solutions were feasible on the original instance. On the R/RC geometries the pipeline beats full-instance savings on distance *and* fleet size. Honest reading: at this fleet size clusters average < 2 super-nodes, so in 3/4 cases the best solution came from the warm-start clustering rather than QUBO sampling — the QUBO stages earn their keep at Experiment-A scale, not here.

C — encoding ablation. Open TSPs, $n = 6..14$ from C101 subsets, 3 trials each, equal budget (Fig. 1).

Under classical SA one-hot never produces an invalid sample while domain-wall degrades from ∼99.7% (n=6) to ∼60% (n=14) valid. This does *not* contradict the literature's pro-domain-wall results (Chen–Stollenwerk–Chancellor 2021): that advantage is a *hardware-embedding* effect — the 1-D wall constraints embed with much shorter chains than one-hot's cliques — and SA has no embedding. It does mean the encoding choice must be re-measured on the QPU (valid rate + chain-break fraction) before trusting either; that is the first thing `run_qpu.py --exp ablation --backend qpu` measures.

5 Takeaways and next steps

What survives scrutiny: (i) TW structure can be pushed into both QUBO stages with zero extra qubits, and the decomposed pipeline stays TW-feasible at $N = 10$ where FQS collapses, reaching a 0.2% gap to the exact optimum where one is known (R101-N5); (ii) the exact coarsening rule's no-repair guarantee held on every full-size run; (iii) the decomposition cuts the largest BQM by 1–3 orders of magnitude. What does not (yet): domain-wall's variable saving costs valid-sample rate under SA, and at Solomon-100 scale the classical warm start usually wins. Next, in priority order: QPU runs of the ablation (settles the encoding question with chain-break data), QPU runs of Experiment A at $N = 10$ (where FQS barely embeds and the decomposed BQMs embed trivially), a penalty-weight sensitivity sweep (all weights here are scaled heuristics), and a Benders-style loop feeding realized route costs back into the assignment QUBO to close the Exp-B gap to the warm start.